

# Задачи по высшей алгебре с РМО

1. (1995) Let  $(M, \cdot)$  be a monoid with identity element  $e$ .
  - (a) Prove that if  $M$  is a finite set, then there are no  $a, b \in M$  such that  $a \cdot b = e$  and  $b \cdot a \neq e$ .
  - (b) Provide an example of functions  $f, g : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}$  such that  $f \circ g = \text{Id}_{\mathbb{Z}_{\geq 0}}$  and  $g \circ f \neq \text{Id}_{\mathbb{Z}_{\geq 0}}$ .
  - (c) Let  $a, b \in M$  be such that  $a \cdot b = e$  and  $b \cdot a \neq e$ . Prove that if  $b^n \cdot a^m = b^q \cdot a^p$  for some positive integers  $m, n, p, q$ , then  $n = q$  and  $m = p$ .
2. Let  $K$  be a finite field. Find the set of integers  $n \geq 2$  having the property that any  $n$ -th degree polynomial in  $K[X]$  with no roots in  $K$  is irreducible over  $K$ .
3. (1996) Prove that a group  $G$  in which exactly two elements other than the identity commute with each other is isomorphic to  $\mathbb{Z}/3\mathbb{Z}$  or  $S_3$ .
4. Let  $A$  be a commutative ring with  $0 \neq 1$  such that for any  $x \in A \setminus \{0\}$  there exist positive integers  $m, n$  such that

$$(x^m + 1)^n = 0.$$

Prove that any endomorphism of  $A$  is an automorphism.

5. (1997) Let  $\alpha \in \mathbb{C} \setminus \mathbb{Q}$  be such that the set

$$A = \{a + b\alpha \mid a, b \in \mathbb{Z}\}$$

is a ring with respect to the usual operations of  $\mathbb{C}$ . If the ring  $A$  has exactly four invertible elements, prove that  $A = \mathbb{Z}[i]$ .

6. Let  $K$  be a finite field,  $n > 2$  an integer,  $f \in K[X]$  an irreducible polynomial of degree  $n$ , and  $g$  the product of all the nonconstant polynomials in  $K[X]$  of degree at most  $n - 1$ . Prove that  $f$  divides  $g - 1$ .
7. (1998)

- (a) Let  $p > 2$  be a natural number and

$$G_p = \bigcup_{n \in \mathbb{N}} \{z \in \mathbb{C} \mid z^{p^n} = 1\}.$$

Prove that  $(G_p, \cdot)$  is a subgroup of  $(\mathbb{C}^*, \cdot)$ .

- (b) Let  $(H, \cdot)$  be an infinite subgroup of  $(\mathbb{C}^*, \cdot)$ . Prove that all proper subgroups of  $H$  are finite if and only if  $H = G_p$  for some prime  $p$ .

8. A ring  $A$  is called *Boolean* if  $x^2 = x$  for every  $x \in A$ . Prove that:

- (a) A finite set  $A$  with  $n \geq 2$  elements can be equipped with the structure of a Boolean ring if and only if  $n = 2^k$  for some positive integer  $k$ .

(b) The set of nonnegative integers can be equipped with the structure of a Boolean ring.

9. Let  $K \subseteq \mathbb{C}$  be a field with the operations inherited from  $\mathbb{C}$  such that

(i)  $K$  has exactly two endomorphisms, say  $f$  and  $g$ ;

(ii) if  $f(x) = g(x)$ , then  $x \in \mathbb{Q}$ .

Prove that there exists a square-free integer  $d \neq 1$  such that  $K = \mathbb{Q}(\sqrt{d})$ .

10. (1999) For a finite group  $G$  we denote by  $n(G)$  the number of elements of the group and by  $s(G)$  the number of subgroups of it. Decide whether the following statements are true or false.

(a) For every  $a > 0$  there exists a finite group  $G$  with  $\frac{n(G)}{s(G)} < a$ .

(b) For every  $a > 0$  there exists a finite group  $G$  with  $\frac{n(G)}{s(G)} > a$ .

11. Let  $A$  be an integral domain and  $A[X]$  be its associated ring of polynomials. For every integer  $n \geq 2$  we define the map

$$\varphi_n : A[X] \rightarrow A[X], \quad \varphi_n(f) = f^n$$

and we assume that the set

$$M = \{ n \in \mathbb{Z}_{\geq 2} : \varphi_n \text{ is an endomorphism of the ring } A[X] \}$$

is nonempty.

Prove that there exists a unique prime number  $p$  such that

$$M = \{ p, p^2, p^3, \dots \}.$$

12. (2000) We say that an abelian group  $G$  has property (P) if, for any commutative group  $H$ , any subgroup  $H' \leq H$  and any monomorphism

$$\mu' : H' \longrightarrow G,$$

there exists a morphism

$$\mu : H \longrightarrow G$$

such that

$$\mu|_{H'} = \mu'.$$

Show that:

(a) the group  $(\mathbb{Q}^*, \cdot)$  does not have property (P);

(b) the group  $(\mathbb{Q}, +)$  has property (P).

13. Prove that a nontrivial finite ring is not a skew field if and only if the equation  $x^n + y^n = z^n$  has nontrivial solutions in this ring for every natural number  $n$ .

14. (2001)

- (a) Consider the polynomial  $P(X) = X^5 \in \mathbb{R}[X]$ . Show that for every  $\alpha \in \mathbb{R}^*$ , the polynomial  $P(X + \alpha) - P(X)$  has no real roots.
- (b) Let  $P(X) \in \mathbb{R}[X]$  be a polynomial of degree  $n \geq 2$ , with real and distinct roots. Show that there exists  $\alpha \in \mathbb{Q}^*$  such that the polynomial  $P(X + \alpha) - P(X)$  has only real roots.

15. Let  $A$  be a finite ring. Show that there exist two natural numbers  $m, p$  with  $m > p \geq 1$ , such that  $a^m = a^p$  for all  $a \in A$ .

16. (2002) Let  $A$  be a ring.

- (a) Show that the set  $Z(A) = \{a \in A \mid ax = xa \text{ for all } x \in A\}$  is a subring of the ring  $A$ .
- (b) Prove that, if any commutative subring of  $A$  is a field, then  $A$  is a field.

17. Let  $K$  be a field having  $q = p^n$  elements, where  $p$  is a prime and  $n > 2$  is an arbitrary integer. For any  $a \in K$ , define the polynomial  $f_a = X^q - X + a$ . Show that:

- (a)  $f = (X^q - X)^q - (X^q - X)$  is divisible by  $f_a$ .
- (b)  $f_a$  has at least  $p^{n-1}$  essentially different irreducible factors in  $K[X]$ .

18. (2003)

- (a) Determine the center of the ring of square matrices of a given dimension with entries in a field  $K$ , and prove that it is isomorphic to  $K$ .
- (b) Prove that

$$(M_n(\mathbb{R}), +, \cdot) \not\cong (M_n(\mathbb{C}), +, \cdot),$$

for any natural number  $n \geq 2$ .

19. For a finite additive group  $L$ , let  $i(L)$  denote the number of multiplicative binary operations on the underlying set of  $L$  such that, together with the given addition, it becomes a ring structure. Prove that:

- (a)  $i(\mathbb{Z}_{12}) = 4$ .
- (b)  $i(A \times B) \geq i(A)i(B)$  for any two finite commutative groups  $A$  and  $B$ .
- (c) There exist two sequences  $(G_k)_{k \geq 1}$ ,  $(H_k)_{k \geq 1}$  of finite commutative groups such that

$$\lim_{k \rightarrow \infty} \frac{\#G_k}{i(G_k)} = 0 \quad \text{and} \quad \lim_{k \rightarrow \infty} \frac{\#H_k}{i(H_k)} = \infty.$$

20. (2004) Let  $f \in \mathbb{Z}[X]$ . For an  $n \in \mathbb{N}$ ,  $n > 2$ , we define  $f_n : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$  by  $f_n(\bar{x}) = \overline{f(x)}$ , for all  $x \in \mathbb{Z}$ .

- (a) Prove that  $f_n$  is well defined.
- (b) Find all polynomials  $f \in \mathbb{Z}[X]$  such that for all  $n \in \mathbb{N}$ ,  $n > 2$ , the function  $f_n$  is surjective.

21. Let  $\mathcal{K}$  be a field of characteristic  $p$ ,  $p \equiv 1 \pmod{4}$ .

- (a) Prove that  $-1$  is the square of an element from  $\mathcal{K}$ .
- (b) Prove that any element  $\neq 0$  from  $\mathcal{K}$  can be written as the sum of three squares, each  $\neq 0$ , of elements from  $\mathcal{K}$ .
- (c) Can  $0$  be written in the same way?

22. (2005) Let  $G$  be a group with  $m$  elements and let  $H$  be a proper subgroup of  $G$  with  $n$  elements. For each  $x \in G$  we denote

$$H^x = \{xhx^{-1} \mid h \in H\}$$

and we suppose that

$$H^x \cap H = \{e\},$$

for all  $x \in G \setminus H$  (where by  $e$  we denote the neutral element of the group  $G$ ).

- (a) Prove that

$$H^x = H^y$$

if and only if

$$x^{-1}y \in H.$$

- (b) Find the number of elements of the set

$$\bigcup_{x \in G} H^x$$

as a function of  $m$  and  $n$ .

23. Let  $A$  be a ring with  $2^n + 1$  elements, where  $n$  is a positive integer, and let

$$M = \{k \in \mathbb{Z} \mid k \geq 2, x^k = x, \forall x \in A\}.$$

Prove that the following statements are equivalent:

- (a)  $A$  is a field;
- (b)  $M$  is nonempty and the smallest element in  $M$  is  $2^n + 1$ .

24. (2006) Let  $K$  be a finite field. Prove that the following statements are equivalent:

- (a)  $1 + 1 = 0$ ;
- (b) for every  $f \in K[X]$  with  $\deg f > 1$ , the polynomial  $f(X^2)$  is reducible.

25. Let  $G$  be a finite group of  $n$  elements ( $n > 2$ ) and let  $p$  be the smallest prime factor of  $n$ . If  $G$  has a unique subgroup  $H$  with  $p$  elements, prove that  $H$  is contained in the center of  $G$ .

*Note.* The center of  $G$  is the set  $Z(G) = \{a \in G \mid ax = xa \text{ for all } x \in G\}$ .

26. (2007) Let  $n \geq 1$  be an integer. Find all rings  $(A, +, \cdot)$  such that for all  $x \in A \setminus \{0\}$  we have  $x^{2^n+1} = 1$ .

27. Let  $n \geq 3$  be an integer and  $S_n$  the permutation group. Let  $G$  be a subgroup of  $S_n$  generated by  $n-2$  transpositions. For every  $k \in \{1, 2, \dots, n\}$ , denote by  $S(k)$  the set

$$\{\sigma(k) : \sigma \in G\}.$$

Show that for any  $k$ ,  $|S(k)| \leq n-1$ .

28. (2008) Let  $A$  be a finite unitary ring with  $n$  elements such that the equation  $x^n = 1$  has a unique solution in  $A$ , namely  $x = 1$ . Prove that

- (a) 0 is the only nilpotent element of  $A$ ;
- (b) there exists an integer  $k \geq 2$  such that the equation  $x^k = x$  has  $n$  solutions in  $A$ .

29. Let  $\mathcal{G}$  be the set of all finite groups with at least two elements.

- (a) Prove that if  $G \in \mathcal{G}$ , then the number of endomorphisms  $f : G \rightarrow G$  is at most  $p^{n/p}$ , where  $p$  is the largest prime divisor of  $n$ , and  $n$  is the number of elements of  $G$ .
- (b) Find all groups in  $\mathcal{G}$  for which the inequality in part (a) is an equality.

30. (2009) Let  $R$  be a finite commutative ring.

- (a) Show that the set of nilpotent elements of  $R$  is closed under both ring operations (addition and multiplication).
- (b) Prove that the number of nilpotent elements of  $R$  divides the number of zero divisors of  $R$ .

31. Find all natural numbers  $n \geq 2$  with the property that the ring  $\mathbb{Z}/n\mathbb{Z}$  has exactly one element that cannot be expressed as a sum of two squares.

32. (2010) We say that a ring  $A$  has property  $(P)$  if any non-zero element can be written uniquely as the sum of an invertible element and a non-invertible element.

- (a) If in  $A$  we have  $1 + 1 = 0$ , prove that  $A$  has property  $(P)$  if and only if  $A$  is a field.
- (b) Give an example of a ring that is not a field, containing at least two elements, and having property  $(P)$ .

33. Let  $G$  be a finite group of order  $n$ . Define the set

$$H = \{x \in G \mid x^2 = e\},$$

where  $e$  is the neutral element of  $G$ . Let  $p = |H|$  be the cardinality of  $H$ . Prove that

- (a)  $|H \cap xH| \geq 2p - n$ , for any  $x \in G$ , where  $xH = \{xh \mid h \in H\}$ .
- (b) If  $p > \frac{3n}{4}$ , then  $G$  is commutative.
- (c) If  $\frac{n}{2} < p \leq \frac{3n}{4}$ , then  $G$  is non-commutative.

34. (2011) Prove that a ring that has a prime characteristic admits nonzero nilpotent elements if and only if its characteristic divides the number of its units.

35. The equation  $z^n + l + - 0$  admits 0 and 1 as its unique solutions in a ring of order  $n > 2$ . Prove that this ring is a skew field.

36. (2012) Let  $(R, +, \cdot)$  be a ring and let  $f$  be a surjective endomorphism of  $R$  such that  $[x, f(x)] = 0$  for any  $x \in R$ , where  $[a, b] = ab - ba$ ,  $a, b \in R$ . Prove that:

- a)  $[x, f(y)] = [f(x), y]$  and  $x[x, y] = f(x)[x, y]$ , for any  $x, y \in R$ ;
- b) If  $R$  is a division ring and  $f$  is different from the identity function, then  $R$  is commutative.

37. Let  $m$  and  $n$  be two nonzero natural numbers. Determine the minimum number of distinct complex roots of the polynomial

$$\prod_{k=1}^n (f + k),$$

when  $f$  covers the set of  $m$ -th degree polynomials with complex coefficients.

38. (2013) Given a ring  $(A, +, \cdot)$  that meets both of the following conditions:

- (1)  $A$  is not a field, and
- (2) For every non-invertible element  $x$  of  $A$ , there is an integer  $m > 1$  (depending on  $x$ ) such that

$$x = x^2 + x^3 + \dots + x^{2^m}.$$

Show that

- (a)  $x + x = 0$  for every  $x \in A$ , and
- (b)  $x^2 = x$  for every non-invertible  $x \in A$ .

39. Given  $n > 2$  a natural number,  $(K, +, \cdot)$  a commutative field such that

$$1 + 1 + \dots + 1 \neq 0 \quad \text{for } m = 2, \dots, n,$$

$f \in K[X]$  a polynomial of degree  $n$  and  $G$  a subgroup of the additive group  $(K[X], +)$ , with  $G \neq K[X]$ . Show that there exists  $a \in K$  such that  $f(a) \notin G$ .

40. (2014) For a ring  $A$ , and an element  $a$  of it, define  $s_a, d_a : A \rightarrow A$  by  $s_a(x) = ax$ ,  $d_a(x) = xa$ .

- (a) Prove that if  $A$  is finite, then  $s_a$  is injective if and only if  $d_a$  is injective.
- (b) Give an example of a ring which has an element  $b$  for which  $s_b$  is injective and  $d_b$  is not, or, conversely,  $s_b$  is not injective, but  $d_b$  is.

41. Let  $G$  be a finite group that has an element  $a \neq 1$  for which exists a prime number  $p$  such that

$$a^p = a^{-1}za \quad \text{for all } z \in G.$$

- (a) Prove that the order of  $G$  is a power of  $p$ .

- (b) Show that  $H := \{x \in G \mid \text{ord}(x) = p\} < G$  and  $\text{ord}(H) > \text{ord}(G)$ .
42. (2015) Let  $R$  be a ring that has the property that all its elements are the product of two idempotent elements of it. Show that:
- a) 1 is the only unit of this ring.
  - b) this ring is Boolean.
43. Show that the set of all elements minus 0 of a finite division ring that has at least 4 elements can be partitioned into two nonempty sets  $A, B$  having the property that

$$\sum_{x \in A} x = \prod_{y \in B} y.$$

44. (2016) Let  $A$  be a ring and let  $D$  be the set of its non-invertible elements. If  $a^2 = 0$  for any  $a \in D$ , prove that:
- a)  $axa = 0$  for all  $a \in D$  and all  $x \in A$ ;
  - b) if  $D$  is a finite set with at least two elements, then there is  $a \in D$ ,  $a \neq 0$ , such that  $ab = ba = 0$ , for every  $b \in D$ .
45. Let  $K$  be a finite field with  $q$  elements,  $q > 3$ . We denote by  $M$  the set of polynomials in  $K[X]$  of degree  $q - 2$  whose coefficients are nonzero and pairwise distinct. Find the number of polynomials in  $M$  that have  $q - 2$  distinct roots in  $K$ .
46. (2017) Let  $n$  be a natural number and  $2n$  real numbers  $b_1, b_2, \dots, b_n$ ,  $a_1 < a_2 < \dots < a_n$ . Show that
- a) if  $b_1, b_2, \dots, b_n > 0$ , then there exists a polynomial  $f \in \mathbb{R}[X]$  irreducible in  $\mathbb{R}[X]$  such that  $f(a_i) = b_i$  for all  $i \in \{1, 2, \dots, n\}$ ;
  - b) there exists a polynomial  $g \in \mathbb{R}[X]$  of degree at least 1 which has only real roots and such that  $g(a_i) = b_i$  for all  $i \in \{1, 2, \dots, n\}$ .
47. Let  $G$  be a finite group with the following property: If  $f$  is an automorphism of  $G$ , then there exists  $m \in \mathbb{N}^*$  so that  $f(x) = x^m$  for all  $x \in G$ . Prove that  $G$  is commutative.
48. (2018) Let  $A$  be a finite ring and  $a, b \in A$ , such that  $(ab - 1)b = 0$ . Prove that  $b(ab - 1) = 0$ .
49. For any  $k \in \mathbb{Z}$ , define

$$F_k = X^4 + 2(1 - k)X^2 + (1 + k).$$

Find all values  $k \in \mathbb{Z}$  such that  $F_k$  is irreducible over  $\mathbb{Z}$  and reducible over  $\mathbb{Z}_p$  for any prime  $p$ .

50. (2019) Let  $n > 4$  be an even natural number and  $G$  be a subgroup of  $\text{GL}_2(\mathbb{C})$  with  $|G| = n$ . Prove that there exists  $H < G$  such that  $\{I_2\} \neq H$  and  $H \neq G$  such that  $XYX^{-1} \in H$ , for all  $X \in G$  and for all  $Y \in H$ .
51. Let  $n > 3$  and  $a_1, a_2, \dots, a_n$  be complex numbers different from 0 with  $|a_i| < 1$  for all  $i \in \{1, 2, \dots, n\}$ . If the coefficients of  $f = \prod_{i=1}^n (X - a_i)$  are integers, prove that
- a) The numbers  $a_1, a_2, \dots, a_n$  are distinct.

b) If  $a_s = a_j a_k$ , then  $s = j = k$ .

52. (2021) Determine all non-trivial finite rings with a unit element in which the sum of all elements is invertible.

53. (2022) Determine all rings  $(A, +, \cdot)$  such that  $a^2 \in \{0, 1\}$  for any  $a \in A$ .

54. Let  $(R, +, \cdot)$  be a ring with center  $Z = \{a \in R : ar = ra, \forall r \in R\}$  with the property that the group  $U = U(R)$  of its invertible elements is finite. Given that  $G$  is the group of automorphisms of the additive group  $(R, +)$ , prove that

$$|G| \geq \frac{|U|}{|Z|}.$$

55. (2023) Let  $(G, \cdot)$  be a finite group with order  $n \in \mathbb{N}^*$ , where  $n \geq 2$ . We say that group  $(G, \cdot)$  is *arrangeable* if there is an ordering of its elements, such that

$$G = \{a_1, a_2, \dots, a_n\} = \{a_1 \cdot a_2, a_2 \cdot a_3, \dots, a_n \cdot a_1\}.$$

a) Determine all positive integers  $n$  for which the group  $(\mathbb{Z}_n, +)$  is arrangeable.

b) Give an example of a group of even order that is arrangeable.

56. Let  $p$  be a prime number,  $n$  a natural number which is not divisible by  $p$ , and  $\mathbb{K}$  a finite field with  $\text{char}(\mathbb{K}) = p$ ,  $|\mathbb{K}| = p^n$ ,  $1_{\mathbb{K}}$  the unity element and  $0_{\mathbb{K}} = 0$ . For every  $m \in \mathbb{N}^*$  we note

$$\hat{m} = \underbrace{1_{\mathbb{K}} + 1_{\mathbb{K}} + \dots + 1_{\mathbb{K}}}_{m \text{ times}},$$

$$f_m = \sum_{k=0}^m (-1)^{m-k} \binom{m}{k} X^{p^k} \in \mathbb{K}[X].$$

a) Show that the roots of  $f_1$  are  $\{\hat{k} \mid k \in \{0, 1, 2, \dots, p-1\}\}$ .

b) Let  $m \in \mathbb{N}^*$ . Determine the set of roots from  $\mathbb{K}$  of polynomial  $f_m$ .

57. (2024) Let  $(\mathbb{K}, +, \cdot)$  be a division ring in which  $x^2 y = y x^2$  for all  $x, y \in \mathbb{K}$ . Prove that  $(\mathbb{K}, +, \cdot)$  is commutative.

58. (2025) We say a ring  $(A, +, \cdot)$  has property  $(P)$  if:

- the set  $A$  has at least 4 elements,
- the element  $1 + 1$  is invertible,
- $x + x^4 = x^2 + x^3$  holds for all  $x \in A$ .

a) Prove that if a ring  $(A, +, \cdot)$  has property  $(P)$ , and  $a, b \in A$  are distinct elements, such that  $a$  and  $a + b$  are units, then  $1 + ab$  is also a unit, but  $b$  is not a unit.

b) Provide an example of a ring with property  $(P)$ .

59. Let  $p$  be an odd prime number, and  $k$  be an odd number not divisible by  $p$ . Consider a field  $K$  with  $kp + 1$  elements, and  $A = \{x_1, x_2, \dots, x_{kp}\}$  be the set of elements of  $K^*$  whose order is not  $k$  in the multiplicative group  $(K^*, \cdot)$ . Prove that the polynomial  $P(X) = (X + x_1)(X + x_2) \dots (X + x_{kp})$  has at least  $p$  coefficients equal to 1.

60. (2026) Let  $n$  be a squarefree positive integer and  $G$  be a group of order  $n$ . Let  $d$  be the largest order of an element of  $G$ , and  $N$  be the number of element of order  $d$  in  $G$ .
- a) Prove that  $N$  is divisible by  $n/d$ .
  - b) Prove that  $N^2 \geq \varphi(n)$ .
61. Let  $p$  be a prime number, and  $A$  be a ring of characteristic  $p$ . Show that  $A$  has at most  $\frac{|A|}{p}$  nilpotent elements.